

MAXIMUM-MODULUS POINTS ON CONCENTRIC CIRCLES

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ABSTRACT. Let f be a nonzero, non-monomial entire function, and let $\nu_f(r)$ count the points at which f attains its maximum modulus on $|z| = r$. Write $f(z) = z^m(a_0 + a_k z^k + O(z^{k+1}))$, where $a_0 a_k \neq 0$. We prove that $\nu_f(r) \leq 2k$ outside a locally finite subset of $(0, \infty)$. The constant $2k$ is sharp. In particular, $\liminf_{r \rightarrow \infty} \nu_f(r) < \infty$, so the number of maximum-modulus points of a fixed non-monomial entire function cannot tend to infinity on all sufficiently large circles. The proof associates the logarithmic derivative $q = zf'/f$ with its reflection $q^*(w) = \overline{q(\bar{w})}$ through the correspondence $q(z) = q^*(w)$ and the map $(z, w) \mapsto (zw, q(z))$. Local valence at the origin bounds the number of inverse pairs that can escape. A local elimination argument then bounds the generic degree of each nonvertical image component. The residue structure of f'/f excludes vertical components.

1. INTRODUCTION

For a nonzero entire function f , let

$$M_f(r) = \max_{|z|=r} |f(z)|, \quad \nu_f(r) = \#\{z : |z| = r, |f(z)| = M_f(r)\}, \quad r > 0.$$

If $f(z) = cz^m$, with $c \neq 0$ and $m \geq 0$, every point of each circle is a maximum-modulus point. Throughout this paper, a non-monomial entire function means a nonzero entire function that is not of this form. For such a function, $\nu_f(r)$ is finite for every $r > 0$, as proved in Lemma 3.2.

A question of Erdős, recorded in Hayman's problem collection [3], distinguishes two possibilities for a single non-monomial entire function: unbounded values of ν_f along a sequence of radii, and divergence of $\nu_f(r)$ as $r \rightarrow \infty$. The all-radii version has also been attributed to Clunie; see the historical discussion in [2]. Herzog and Piranian [4] constructed an entire function with $\nu_f(n) = n$ for every positive integer n . Pardo-Simón and Sixsmith [7] subsequently obtained unbounded selected-radius multiplicity for a finite-order entire function in the Eremenko–Lyubich class $\mathcal{B}$. Neither assertion controls the intervening radii.

Glücksam and Pardo-Simón [2] proved an approximate all-radii result: for every fixed $\varepsilon > 0$, the number of components of

$$\{z : |z| = r, |f(z)| > M_f(r) - \varepsilon\}$$

tends to infinity for one entire function. The components need not all contain exact maximum-modulus points. Marchenko [6, p. 215, Conjecture 1] conjectured that $\liminf_{r \rightarrow \infty} \nu_f(r) < \infty$ for every non-monomial entire function of finite lower order. We prove this conclusion without a growth assumption, together with a sharp quantitative bound.

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Theorem 1.1. *Let f be a non-monomial entire function. Write*

$$f(z) = z^m(a_0 + a_k z^k + O(z^{k+1})), \quad m \geq 0, \quad k \geq 1, \quad a_0 a_k \neq 0. \quad (1.1)$$

Then the set

$$E_f = \{r > 0 : \nu_f(r) > 2k\}$$

is locally finite in $(0, \infty)$. The constant $2k$ is sharp.

Here $m = \text{ord}_0 f$ and $k = \text{ord}_0(g - g(0))$, where $g = z^{-m}f$. Thus k is the first positive exponent occurring in g . This definition differs from the inner degree, which is the greatest common divisor of the positive exponents occurring in g ; the two integers can coincide. See [1]. Local finiteness means that E_f meets every compact subinterval of $(0, \infty)$ in finitely many points.

Corollary 1.2. *For every non-monomial entire function satisfying (1.1),*

$$\liminf_{r \rightarrow \infty} \nu_f(r) \leq 2k.$$

In particular, no non-monomial entire function satisfies $\nu_f(r) \rightarrow \infty$ as $r \rightarrow \infty$.

Proof. Each interval $(n, n+1)$ contains a radius outside E_f . Choosing one such radius for every positive integer n proves the assertion. $\square$

Local finiteness permits an unbounded sequence of exceptional radii. Thus Theorem 1.1 is consistent with the selected-radius constructions above. Section 4 gives elementary examples showing both sharpness of the bound and the necessity of allowing exceptional radii.

Outline of the proof. Set $q = zf'/f$ and $q^*(w) = \overline{q(\bar{w})}$. At a differentiability point of $H(x) = \log M_f(e^x)$, all maxima on the circle $|z| = e^x$ satisfy $q(z) = H'(x) \in \mathbb{R}$. They therefore determine inverse pairs of

$$(z, w) \mapsto (zw, q(z)), \quad q(z) = q^*(w),$$

over the single point $(e^{2x}, H'(x))$. This map is proper away from the value $q(0)$ in its second coordinate. Its image is a complex-analytic curve, constructed locally from finite products of inverse roots.

A component whose generic fiber has more than $2k$ pairs cannot lose all its pairs through zero or infinity as the second coordinate tends to $q(0)$. At least one bounded pair survives. This supplies a local polynomial equation across the omitted value, and the image component extends analytically. Its projection to the product coordinate is proper. Unless the component is vertical, it must reach arbitrarily small nonzero products, where local valence gives at most $2k$ pairs. Vertical components are incompatible with the positive residues of f'/f . This proves the bound outside a null set of radii. A local parametrization of stationary arcs then makes the exceptional set locally finite.

2. A MULTIPLICATIVE CORRESPONDENCE

We first separate the analytic argument from the entire-function application. Let A and B be nonconstant meromorphic functions on $\mathbb{C}$, holomorphic at zero, such that

$$A(0) = B(0) = c \in \mathbb{C}, \quad k_A = \text{ord}_0(A - c), \quad k_B = \text{ord}_0(B - c). \quad (2.1)$$

Let $L = k_A + k_B$, and write $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$. All equalities of meromorphic functions below are understood as equalities of maps to the Riemann sphere $\hat{\mathbb{C}}$. An inverse pair over (s, p) means an ordered pair $(z, w) \in (\mathbb{C}^*)^2$ with $zw = s$ and $A(z) = B(w) = p$. All fiber cardinalities count distinct inverse pairs; root multiplicities enter only in the local polynomials and mapping degrees.

Consider the reduced analytic curve

$$\mathcal{C} = \{(z, w) \in (\mathbb{C}^*)^2 : A(z) = B(w)\},$$

and the holomorphic map

$$\Phi : \mathcal{C} \longrightarrow \mathbb{C}^* \times \widehat{\mathbb{C}}, \quad \Phi(z, w) = (zw, A(z)). \quad (2.2)$$

Put

$$\Omega = \mathbb{C}^* \times (\widehat{\mathbb{C}} \setminus \{c\}), \quad \mathcal{C}^\circ = \Phi^{-1}(\Omega), \quad Y = \Phi(\mathcal{C}^\circ).$$

The source has pure complex dimension one. Indeed, near a pair (z_0, w_0) with finite common value p_0 , write

$$A(z) - p_0 = (z - z_0)^a h_A(z), \quad B(w) - p_0 = (w - w_0)^b h_B(w),$$

where $a, b \geq 1$ and the two holomorphic factors are nonvanishing. Choosing local holomorphic roots of these factors gives conformal coordinates

$$u = (z - z_0)h_A(z)^{1/a}, \quad v = (w - w_0)h_B(w)^{1/b},$$

in which the correspondence is $u^a = v^b$. If $d = \gcd(a, b)$, its d local branches admit parametrizations

$$u = \zeta t^{b/d}, \quad v = t^{a/d}, \quad \zeta^{a/d} = \omega, \quad \omega^d = 1.$$

One choice of ζ for each ω suffices. These are one-to-one parametrizations of the respective curve germs, because a/d and b/d are coprime. At a common pole, apply the same argument to $1/A$ and $1/B$, whose common value is zero.

Lemma 2.1 (Small-coordinate fibers). *There exists $\varepsilon > 0$ such that, for every $(s, p) \in \mathbb{C}^* \times \widehat{\mathbb{C}}$, at most L pairs satisfying*

$$zw = s, \quad A(z) = B(w) = p \quad (2.3)$$

have $\min(|z|, |w|) < \varepsilon$. In particular, if $0 < |s| < \varepsilon^2$, the whole fiber has at most L distinct pairs. We may also choose ε so that neither A nor B takes the value c in $0 < |z| < \varepsilon$.

Proof. Write $A(z) - c = z^{k_A} h_A(z)$ with $h_A(0) \neq 0$, and choose a holomorphic k_A th root of h_A in a small disk. Then

$$\tau_A(z) = z h_A(z)^{1/k_A}, \quad \tau'_A(0) = h_A(0)^{1/k_A} \neq 0, \quad A(z) - c = \tau_A(z)^{k_A}.$$

Shrink the disk until τ_A is injective and A has no poles there. For finite p , the equation $\tau_A(z)^{k_A} = p - c$ has at most k_A distinct solutions, even if some k_A th roots of $p - c$ lie outside the coordinate image. For $p = \infty$ there are no solutions in the disk. Repeat this for B , and take a common disk $|z| < \varepsilon$ on which both conclusions hold. For fixed $s \neq 0$, choosing either coordinate determines the other. There are therefore at most k_A pairs with small first coordinate and at most k_B with small second coordinate. If $|s| < \varepsilon^2$, every pair has a small coordinate. The assertion about the value c follows from the same local coordinates. $\square$

2.1. Local product equations. The elementary elimination fact used in both the image and extension arguments is stated explicitly.

Lemma 2.2 (Local product polynomial). *Let A and B be holomorphic on neighborhoods of finite unions of closed, pairwise disjoint disks U_i and V_j , respectively. Suppose $A - p_0$ and $B - p_0$ have no zeros on the disk boundaries. List the roots of $A(z) = p$ in the U_i as $\alpha_1(p), \dots, \alpha_N(p)$, and the roots of $B(w) = p$ in the V_j as $\beta_1(p), \dots, \beta_M(p)$, with multiplicity. For p near p_0 ,*

$$P(s, p) = \prod_{i=1}^N \prod_{j=1}^M (s - \alpha_i(p)\beta_j(p)) \quad (2.4)$$

is a monic polynomial in s with holomorphic coefficients in p . Its zero set consists exactly of the products of these local inverse roots. In particular, $P(s, p_0)$ is not identically zero as a polynomial in s .

Proof. If either union of disks is empty, the assertion holds with $P = 1$. Assume that both unions are nonempty. The minimum of $|A - p_0|$ over the finitely many boundaries ∂U_i is positive, and similarly for $B - p_0$ on the ∂V_j . Choose a parameter disk smaller than both minima. Rouché's theorem makes the total root counts N and M constant on that disk. If either count is zero, then again $P = 1$; hence assume $NM \geq 1$. For $n \geq 1$, the argument principle gives holomorphic power sums

$$u_n(p) = \frac{1}{2\pi i} \sum_i \int_{\partial U_i} \frac{z^n A'(z)}{A(z) - p} dz, \quad v_n(p) = \frac{1}{2\pi i} \sum_j \int_{\partial V_j} \frac{w^n B'(w)}{B(w) - p} dw.$$

The denominators stay nonzero on the fixed contours, so the integrals depend holomorphically on p . The residue at a root α of multiplicity r is $r\alpha^n$. Consequently,

$$u_n(p) = \sum_{i=1}^N \alpha_i(p)^n, \quad v_n(p) = \sum_{j=1}^M \beta_j(p)^n,$$

without any need to choose single-valued inverse branches. The n th power sum of the NM product roots is

$$T_n(p) = \sum_{i=1}^N \sum_{j=1}^M (\alpha_i(p)\beta_j(p))^n = u_n(p)v_n(p).$$

Put $n_P = NM$ and write

$$P(s, p) = s^{n_P} + c_1(p)s^{n_P-1} + \cdots + c_{n_P}(p), \quad c_0(p) = 1.$$

Newton's identities give the explicit recursion

$$\ell c_\ell(p) + \sum_{n=1}^{\ell} c_{\ell-n}(p)T_n(p) = 0, \quad 1 \leq \ell \leq n_P. \quad (2.5)$$

Induction on ℓ proves that every c_ℓ is holomorphic. For example, $c_1 = -T_1$ when $n_P \geq 1$, and $c_2 = (T_1^2 - T_2)/2$ when $n_P \geq 2$. At each fixed p , the same identities recover the coefficients of the actual finite product in (2.4). Thus its roots are exactly the products of the local inverse roots, including at coalescence values. At $p = p_0$ the polynomial specializes to

$$P(s, p_0) = \prod_{\alpha, \beta} (s - \alpha\beta)^{\text{ord}_\alpha(A-p_0) \text{ord}_\beta(B-p_0)},$$

where the product runs over the central roots in the chosen disks. This polynomial is monic, so it cannot vanish identically in s . $\square$

Individual inverse roots in Lemma 2.2 may ramify; no single-valued labeling is asserted. If one collection is empty, the empty product is $P = 1$. Near a common pole use the holomorphic functions $1/A$, $1/B$ and the parameter $\lambda = 1/p$. Their local root products give $P(s, \lambda)$ by exactly the same recursion.

Lemma 2.3 (Properness and the analytic image). *The map $\Phi : \mathcal{C}^\circ \rightarrow \Omega$ is proper with finite fibers. Its image Y , with the reduced structure, is a closed pure one-dimensional analytic subset of Ω .*

Proof. Let $K \subset \Omega$ be compact. Its p -projection is a compact subset of $\widehat{\mathcal{C}}$ avoiding c . Continuity of A and B at zero gives $\eta > 0$ such that every inverse pair over K has $|z|, |w| \geq \eta$. If $|s| \leq S$ on K , then

$$\eta \leq |z|, |w| \leq S/\eta. \quad (2.6)$$

The inverse image of K is closed in this compact annulus product, and is therefore compact. For a fixed (s, p) , the equation $A(z) = p$ has finitely many roots in that annulus, and $w = s/z$ is determined. This includes $p = \infty$.

Properness makes Y closed. To obtain its local equations, fix $y_0 = (s_0, p_0) \in Y$ and take small disjoint disks around every first and second coordinate occurring in its finite fiber. For finite p_0 the disks avoid poles; for $p_0 = \infty$ use reciprocal value coordinates. Properness implies that every inverse pair over a sufficiently small target neighborhood lies in the products of these disks. Otherwise a sequence of inverse pairs outside them would have a subsequence converging to the central fiber.

Apply Lemma 2.2 to the coordinate disks. Every nearby point of Y satisfies $P(s, p) = 0$. Conversely, each nearby zero of P is the product of actual inverse roots, so it belongs to Y . Thus $Y = \{P = 0\}_{\text{red}}$ locally. The monic polynomial is nonzero and gives a hypersurface in a two-dimensional coordinate neighborhood, hence a pure analytic curve. No source component is contracted, since all fibers are finite. $\square$

2.2. Generic degrees. We use the standard local structure of reduced analytic curves: finite irreducible decomposition, one-to-one disk parametrizations of irreducible curve germs, connected punctured branches, and a locally finite singular set. The regular locus of a globally irreducible curve is connected. These facts, and the local finite-covering statement, can be found in [5, §6.7: Proposition 6.7.3, Theorems 6.7.4 and 6.7.6, and Exercise 6.7.3]. For clarity, a normalization separates the finitely many branches at a singular curve point: each local branch is represented by its own holomorphic disk parameter. The resulting Riemann surface is connected for an irreducible curve.

For an irreducible component Γ of Y , define d_Γ to be the generic number of distinct pairs in a fiber over Γ , including all source components above it.

Lemma 2.4 (Generic degree and smooth fibers). *The integer d_Γ is finite and well defined. If $y \in \Gamma$ is a smooth point of the total reduced image Y , then*

$$\#\Phi^{-1}(y) \leq d_\Gamma. \quad (2.7)$$

Generic points with exactly d_Γ distinct inverse pairs are dense in Γ .

Proof. Fix a smooth point $y_0 \in \Gamma$ of the total image Y , and choose a holomorphic coordinate u on a target disk in Y , with $u(y_0) = 0$. The inverse fiber is finite. Parametrize every local source branch above its points by a disk map $\gamma_\ell(t)$, with $\gamma_\ell(0)$ in that fiber. Each parametrization is one-to-one away from its central singular point. In a sufficiently small neighborhood the only image branch is the target disk. Hence

$$h_\ell(t) = u(\Phi(\gamma_\ell(t)))$$

is holomorphic. It is nonconstant: otherwise a whole source branch would lie in one fiber, contrary to Lemma 2.3. Write

$$h_\ell(t) = a_\ell t^{e_\ell} (1 + tb_\ell(t)), \quad a_\ell \neq 0, \quad e_\ell \geq 1.$$

The change of coordinate

$$v_\ell = t[a_\ell(1 + tb_\ell(t))]^{1/e_\ell}$$

is conformal near zero and gives $h_\ell = v_\ell^{e_\ell}$. Thus this normalized source branch contributes exactly e_ℓ preimages to each sufficiently small nonzero target value, and one normalized preimage to the central value.

Choose disjoint neighborhoods of the distinct source points and small representatives of all their branches. Properness ensures that every inverse pair over a sufficiently small target disk is in these neighborhoods. Indeed, omitted pairs over a sequence tending to y_0 would have a convergent subsequence in the central fiber. No inverse pairs are therefore missing from the local count. For

generic nearby values the branch preimages are also distinct as source points: two distinct local branches intersect only at their center. The total local degree is

$$D(y_0) = \sum_{\ell} e_{\ell}. \quad (2.8)$$

At any point of this target disk the number of normalized preimages, counted without multiplicity, is at most $D(y_0)$. The number of actual source points is no larger, since different normalized points can represent the same singular source point. This proves the local form of (2.7).

The degrees in (2.8) agree on overlapping target disks: both equal the actual fiber cardinality at a generic point of their overlap. The regular locus of Γ is connected. Deleting its intersections with other image components deletes a locally finite set; a connected Riemann surface remains connected after such a deletion. The local degrees therefore give one finite positive integer d_{Γ} on all smooth points of the total image belonging to Γ .

The same local normal forms show that, after shrinking a smooth target disk centered at y_0 , every point other than its center has exactly $D(y_0) = d_{\Gamma}$ distinct inverse pairs. Thus the exceptional fibers form a locally finite set within the smooth locus of Y . The singular locus of Y is also locally finite, and each branch at a singular point has a punctured smooth representative. Generic points are therefore dense in the whole of Γ , as asserted. $\square$

The smoothness requirement in (2.7) concerns the *total* image Y , not merely one component. At a singular image point, pairs belonging to several local image branches may occur in the same fiber. Also, d_{Γ} is a degree of the source-to-image map. The degree NM of a local product polynomial in the variable s counts products of local inverse roots, with multiplicity. These are different quantities, and no equality between them is used.

2.3. Continuation at the omitted value. The following lemma is the main continuation step.

Lemma 2.5 (Continuation across the omitted value). *If $d_{\Gamma} > L$, then the closure of Γ in $\mathbb{C}^* \times \widehat{\mathbb{C}}$ is a closed irreducible analytic curve. Moreover, if*

$$\Sigma = \{\alpha\beta : \alpha, \beta \neq 0, A(\alpha) = B(\beta) = c\},$$

then

$$\overline{\Gamma} \cap (\mathbb{C}^* \times \{c\}) \subset \Sigma \times \{c\}. \quad (2.9)$$

The set Σ is locally finite and contains no s with $0 < |s| < \varepsilon^2$.

Proof. The roots of $A - c$ and $B - c$ in $\mathbb{C}^*$ avoid the disk $|z| < \varepsilon$, by Lemma 2.1. A bound $|\alpha\beta| \leq S$ therefore bounds both root coordinates between ε and S/ε . Only finitely many roots occur in that compact annulus. This proves the assertions about Σ .

Fix a boundary point $y_0 = (s_0, c)$ of Γ , with $s_0 \neq 0$, and choose a target neighborhood on which $|s| \leq S$. Every generic point of Γ in this neighborhood has more than L distinct inverse pairs. At most L have a coordinate smaller than ε . Consequently, at least one pair satisfies

$$\varepsilon \leq |z|, |w| \leq S/\varepsilon. \quad (2.10)$$

A sequence of such pairs over generic points tending to y_0 has a convergent subsequence. Its limit (α, β) satisfies $A(\alpha) = B(\beta) = c$ and $\alpha\beta = s_0$. This proves (2.9). The chosen pair may change from one target point to the next.

We next obtain one analytic equation containing the whole nearby component. There are finitely many roots of $A - c$ and $B - c$ in an enlarged compact annulus containing (2.10). Choose small coordinate disks around all of them. For p sufficiently close to c , every pair satisfying (2.10) has its coordinates in those disks. This follows from compactness and continuity; a contrary sequence would converge to an omitted root of $A - c$ or $B - c$.

Lemma 2.2 now supplies a polynomial $P_0(s, p)$, monic in s and holomorphic in p across c , from the inverse roots in these disks. Every generic point of Γ sufficiently near y_0 satisfies $P_0(s, p) = 0$, because one bounded inverse pair is available. Generic points are dense, and P_0 is continuous, so there is a neighborhood U of y_0 for which

$$\Gamma \cap U \subset Z := \{(s, p) \in U : P_0(s, p) = 0\}_{\text{red}}. \quad (2.11)$$

Only one bounded inverse pair per generic target point was needed for this containment; the other inverse pairs may escape. The central specialization is

$$P_0(s, c) = \prod_{\alpha, \beta} (s - \alpha\beta)^{\text{ord}_\alpha(A-c) \text{ord}_\beta(B-c)}, \quad (2.12)$$

which is not identically zero.

Since $P_0(s, c)$ is a nonzero polynomial, choose a small s -disk around s_0 in which s_0 is its only root and whose boundary contains no root. After shrinking the p -disk around c , P_0 is still nonzero on that s -boundary. The local curve Z thus has finitely many irreducible branches at y_0 , none contained in $p = c$. For each branch choose a one-to-one normalized disk parametrization $\gamma_j(t)$ centered at y_0 . Its p -coordinate satisfies

$$p(\gamma_j(t)) - c = a_j t^{n_j} (1 + O(t)), \quad a_j \neq 0.$$

A holomorphic change of parameter reduces this to $p(\gamma_j(t)) - c = t^{n_j}$. In particular, the branch meets $p = c$ only at its center, and its punctured representative is connected. Shrink the representatives further so that they are smooth and pairwise disjoint away from y_0 .

On each punctured branch Z_j^* , the set $\Gamma \cap Z_j^*$ is closed, because Γ is closed in Ω and $Z_j^* \subset \Omega$. It is also open. To verify this, let $y \in \Gamma \cap Z_j^*$. By (2.11), the germ of Γ at y is a one-dimensional analytic subset of the smooth one-dimensional germ of Z_j^* . In a disk coordinate, a proper analytic subset is discrete; it cannot have dimension one. Hence Γ contains a neighborhood of y in Z_j^* . Connectedness now gives

$$\Gamma \cap Z_j^* = Z_j^* \quad \text{or} \quad \Gamma \cap Z_j^* = \emptyset.$$

It follows that the closure of Γ near y_0 is a union of a subset of the finitely many full local branches of Z , and is analytic. At points of $p = c$ not in the closure there is nothing to extend. This gives a closed analytic curve globally. It remains irreducible because Γ is dense and irreducible. Indeed, a decomposition of the closure into two proper closed analytic subsets would restrict to such a decomposition of Γ : neither subset could contain all of Γ , since it would then contain its closure. No curve component is added inside $p = c$, by (2.12). $\square$

Theorem 2.6 (Degree bound). *For every irreducible component Γ of Y on which the product coordinate s is nonconstant,*

$$d_\Gamma \leq k_A + k_B.$$

Proof. Suppose $d_\Gamma > L$. By Lemma 2.5, the analytic closure $\bar{\Gamma}$ lies in $\mathbb{C}^* \times \hat{\mathbb{C}}$. Its projection

$$\pi : \bar{\Gamma} \longrightarrow \mathbb{C}^*, \quad \pi(s, p) = s,$$

is proper: for compact $K \subset \mathbb{C}^*$, its inverse image is a closed subset of the compact set $K \times \hat{\mathbb{C}}$. In particular, its image is closed.

The projection is also open. Pull s back to the connected normalization of the irreducible curve $\bar{\Gamma}$. It is a holomorphic function there. If s were constant on one local branch, this pullback would be constant on an open disk, and hence constant everywhere by the identity theorem. That would make s constant on Γ , contrary to the hypothesis. Thus s is nonconstant on every parametrizing disk. The open mapping theorem applies at its center even when its derivative vanishes, and shows that the image of every neighborhood of every point contains a neighborhood of that point's s -value. Thus $\pi(\bar{\Gamma})$ is nonempty, open, and closed in $\mathbb{C}^*$, and equals $\mathbb{C}^*$.

In particular, $\bar{\Gamma}$ has points with $0 < |s| < \varepsilon^2$. Such points cannot have $p = c$, by Lemma 2.5, and therefore lie in Γ . This nonempty open part of Γ contains generic points with $d_\Gamma > L$ inverse pairs. Lemma 2.1 bounds every fiber there by L , a contradiction. $\square$

Remark 2.7. The nonconstant-product hypothesis is necessary for general meromorphic functions. For

$$A(z) = B(z) = \frac{z + z^3}{1 + z^4},$$

one has $A(0) = 0$, $A'(0) = 1$, and $A(z) = A(1/z)$. The image component $s = 1$ has four inverse pairs over a generic p , although $k_A + k_B = 2$. The entire-function application excludes these vertical components by a residue argument. Indeed, $A(z) = p$ is equivalent to

$$pz^4 - z^3 - z + p = 0.$$

For a generic nonzero p this quartic has four distinct nonzero roots, producing four distinct pairs $(z, 1/z)$ over $(1, p)$.

3. MAXIMUM-MODULUS POINTS

Let f satisfy (1.1), and set

$$q(z) = \frac{zf'(z)}{f(z)}, \quad q^*(w) = \overline{q(\bar{w})}. \quad (3.1)$$

To compute the expansion explicitly, write $f = z^m g$. Then

$$q(z) = m + \frac{zg'(z)}{g(z)}, \quad g(z) = a_0 + a_k z^k + O(z^{k+1}), \quad zg'(z) = ka_k z^k + O(z^{k+1}).$$

Since $1/g(z) = a_0^{-1} + O(z^k)$, the removable value at zero and the first nonconstant term are

$$q(0) = q^*(0) = m, \quad q(z) = m + \frac{ka_k}{a_0} z^k + O(z^{k+1}). \quad (3.2)$$

Apply Section 2 with $A = q$, $B = q^*$, and $c = m$.

Lemma 3.1 (Exclusion of reciprocal symmetry). *There is no $s_0 \in \mathbb{C}^*$ for which*

$$q(z) = q^*(s_0/z) \quad (z \in \mathbb{C}^*). \quad (3.3)$$

Consequently, no irreducible component of Y has constant product coordinate.

Proof. If (3.3) held, the right side would tend to m at infinity. Hence q would extend holomorphically at infinity and be rational, with $q(\infty) = q(0) = m$. The rational differential

$$\frac{q(z)}{z} dz = \frac{f'(z)}{f(z)} dz$$

has residue m at zero. At infinity, set $\zeta = 1/z$. Then

$$\frac{q(z)}{z} dz = -\frac{q(1/\zeta)}{\zeta} d\zeta,$$

so the residue is $-m$. If $a \neq 0$ is a zero of f of multiplicity n_a , write $f(z) = (z - a)^{n_a} u(z)$ with $u(a) \neq 0$. Locally,

$$\frac{f'(z)}{f(z)} = \frac{n_a}{z - a} + \frac{u'(z)}{u(z)}.$$

Thus every remaining pole has a positive integer residue n_a . The residue theorem on the sphere gives

$$0 = m - m + \sum_{a \neq 0: f(a)=0} n_a.$$

The sum is finite because the differential is rational. Since all its summands are positive, there are no nonzero zeros of f and no other poles of q . The rational function q is therefore holomorphic on the sphere, hence constant m . Finally, $q = m + zg'/g$ for $g = z^{-m}f$ gives $g' = 0$, and $f(z) = Cz^m$, a contradiction.

If an image component had $s = s_0$, a nonconstant source branch above it would satisfy $w = s_0/z$. Its z -coordinate is nonconstant, otherwise both coordinates would be constant. Its image contains an open set of z -values, on which (3.3) holds. The meromorphic identity theorem gives (3.3) on $\mathbb{C}^*$, which has just been excluded. $\square$

Lemma 3.2 (Finiteness on each circle). *A non-monomial entire function has finitely many maximum-modulus points on each circle of positive radius.*

Proof. The function $\theta \mapsto |f(re^{i\theta})|^2$ is real analytic. If its maximum set were infinite, compactness would give an accumulation point of the zeros of $|f(re^{i\theta})|^2 - M_f(r)^2$. The real-analytic identity theorem would then make the modulus constant on the circle. In that case, with $f^*(w) = \overline{f(\bar{w})}$, the identity theorem gives

$$f(z)f^*(r^2/z) = M_f(r)^2 \quad (z \in \mathbb{C}^*).$$

Indeed, the product is holomorphic on $\mathbb{C}^*$ and equals the positive constant $M_f(r)^2$ on the circle, which has accumulation points in that domain. The displayed identity excludes any nonzero zero of f . Thus $g = z^{-m}f$ is a zero-free entire function and has constant modulus $M_f(r)/r^m$ on the circle. The maximum-modulus principle applied to both g and $1/g$ gives, for $|z| < r$,

$$|g(z)| \leq \frac{M_f(r)}{r^m}, \quad |g(z)| \geq \frac{M_f(r)}{r^m}.$$

So g is constant on the disk, hence everywhere, and f is a monomial. This contradiction proves that the original maximum set is finite. $\square$

3.1. The almost-everywhere bound.

Proposition 3.3 (The almost-everywhere bound). *One has $\nu_f(r) \leq 2k$ for Lebesgue-almost every $r > 0$.*

Proof. Put $H(x) = \log M_f(e^x)$. Hadamard's three-circle theorem makes H convex, hence locally Lipschitz and differentiable almost everywhere. Fix a differentiability point x and a global maximum $z = e^{x+i\theta}$. Its value is nonzero, because $M_f(e^x) > 0$. Choose a holomorphic logarithm $F(w) = \log f(e^w)$ near $w = x + i\theta$. Its derivatives satisfy

$$F'(w) = q(e^w), \quad \frac{\partial}{\partial x} \operatorname{Re} F(x + i\theta) = \operatorname{Re} q(z), \quad \frac{\partial}{\partial \theta} \operatorname{Re} F(x + i\theta) = -\operatorname{Im} q(z). \quad (3.4)$$

Angular stationarity gives $\operatorname{Im} q(z) = 0$. On a neighborhood of x , the nonnegative function

$$t \mapsto H(t) - \log |f(e^{t+i\theta})|$$

is differentiable at x and has a local minimum zero there. Its derivative at x therefore vanishes. By (3.4), $H'(x) - \operatorname{Re} q(z) = 0$, so

$$q(z) = H'(x) \in \mathbb{R}. \quad (3.5)$$

Thus each distinct global maximum gives a distinct pair $(z, \bar{z})$ over

$$(s, p) = (e^{2x}, H'(x)). \quad (3.6)$$

Here $z\bar{z} = e^{2x}$ and $q^*(\bar{z}) = \overline{q(z)} = H'(x)$.

These targets stay away from the omitted value $p = m$ on each compact x -interval. Indeed, write $f = z^m g$, where $g(0) \neq 0$ and g is nonconstant. The function $J(x) = \log M_g(e^x)$ is convex and strictly increasing. Strict increase follows from the maximum-modulus principle applied to

two concentric disks: if $M_g(r_1) = M_g(r_2)$ for $0 < r_1 < r_2$, a point on the inner circle attains the maximum inside the outer disk, forcing g to be constant. For a convex function, slopes of successive secants are ordered, and the increments $J(t+1) - J(t)$ are nondecreasing in t . Hence at a differentiability point $x \in [a, b]$,

$$\begin{aligned} J(a) - J(a-1) &\leq J(x) - J(x-1) \leq J'(x) \\ &\leq J(x+1) - J(x) \leq J(b+1) - J(b). \end{aligned}$$

Strict increase makes the first increment positive. We obtain

$$0 < \delta_a := J(a) - J(a-1) \leq J'(x) \leq C_b := J(b+1) - J(b) < \infty. \quad (3.7)$$

Since $H' = m + J'$, the targets in (3.6) lie in the compact set

$$K_{a,b} = [e^{2a}, e^{2b}] \times [m + \delta_a, m + C_b] \subset \Omega.$$

The singular locus of the reduced curve Y is locally finite, so only finitely many singular image points lie in this compact set. Their product coordinates exclude only finitely many radii. At every remaining differentiability radius the target is smooth in the total image, and Lemma 2.4, Theorem 2.6, and Lemma 3.1 give

$$\nu_f(e^x) \leq \#\Phi^{-1}(e^{2x}, H'(x)) \leq d_\Gamma \leq 2k.$$

This excludes only a null set on each compact x -interval. Exponentiation preserves null sets on such intervals, proving the assertion for $r > 0$. $\square$

3.2. Local constancy on either side of a radius. The next elementary local argument turns the almost-everywhere conclusion into the stated exceptional-set theorem. It allows angular degeneracies and does not assume that H is analytic through the central radius.

Lemma 3.4 (One-sided local constancy). *For every $r_0 > 0$, there exists $\delta > 0$ such that ν_f is constant on each of $(r_0 - \delta, r_0)$ and $(r_0, r_0 + \delta)$, with $\delta < r_0$. The two constants need not agree with each other or with $\nu_f(r_0)$.*

Proof. By Lemma 3.2, the central circle has finitely many maxima. We first describe the stationary set near each of them, and then show that these local descriptions cover all nearby maxima.

Fix one central maximum and choose a logarithmic coordinate w near it. Put $Q(w) = q(e^w)$ and let w_0 be the central point, with $x_0 = \operatorname{Re} w_0 = \log r_0$. The function Q is holomorphic near w_0 and nonconstant: constancy there would make q constant everywhere and f a monomial. Angular stationarity gives $p_0 = Q(w_0) \in \mathbb{R}$. Factor

$$Q(w) - p_0 = (w - w_0)^d h(w), \quad d \geq 1, \quad h(w_0) \neq 0.$$

Choosing a local holomorphic d th root of h gives the conformal coordinate $\chi(w) = (w - w_0)h(w)^{1/d}$, in which

$$Q(w) - p_0 = \chi(w)^d, \quad \chi(w_0) = 0.$$

Angular stationarity is $\operatorname{Im} Q(w) = 0$, so its local set consists of the finitely many real-analytic half-arcs

$$w_j(t) = \chi^{-1}(e^{i\pi j/d} t), \quad t \geq 0, \quad j = 0, \dots, 2d-1. \quad (3.8)$$

All half-arcs are restricted to sufficiently small t .

Choose disjoint angular windows $I_1, \dots, I_{n_0}$ about the central maxima so that their thin radial neighborhoods are contained in the coordinate neighborhoods just constructed and avoid zeros of f . Put $M_0 = M_f(r_0)$. Compactness gives $\eta > 0$ such that

$$|f(r_0 e^{i\theta})|^2 \leq M_0^2 - 3\eta \quad \left(\theta \notin \bigcup_{j=1}^{n_0} I_j \right).$$

For r sufficiently close to r_0 , uniform continuity bounds the left side with r_0 replaced by r by $M_0^2 - 2\eta$ on this complement. At any fixed central maximizing angle, the squared modulus is greater than $M_0^2 - \eta$. Every nearby global maximum therefore lies in a window and belongs to one of the stationary half-arcs in (3.8).

For each stationary half-arc, put $x_j(t) = \operatorname{Re} w_j(t)$. This function cannot be identically x_0 . Otherwise the stationary set would contain an open angular arc on $|z| = r_0$. The modulus would be constant on that arc, hence on the circle by real analyticity, contradicting Lemma 3.2. Therefore

$$x_j(t) - x_0 = c_j t^{\ell_j} (1 + ta_j(t)), \quad c_j \in \mathbb{R} \setminus \{0\}, \quad \ell_j \geq 1. \quad (3.9)$$

Here a_j is real analytic near zero. Shrink the parameter interval so that $1 + ta_j(t) > 0$, put $\sigma_j = \operatorname{sign}(c_j)$, and define

$$\rho_j(t) = t[|c_j|(1 + ta_j(t))]^{1/\ell_j}. \quad (3.10)$$

The root is chosen positive for real t near zero. This is a real-analytic function with $\rho_j'(0) = |c_j|^{1/\ell_j} > 0$. The real-analytic inverse function theorem gives its inverse $T_j = \rho_j^{-1}$, and

$$x_j(t) = x_0 + \sigma_j \rho_j(t)^{\ell_j}.$$

Thus the arc lies on the side σ_j for small positive t and intersects each sufficiently close radius on that side once.

Now include all half-arcs from all central windows, relabel them by one finite index set, and choose ℓ_* to be a common multiple of their finitely many exponents ℓ_j . Fix a side $\sigma \in \{-1, 1\}$ and put $x = x_0 + \sigma v^{\ell_*}$ with $v > 0$. Exactly the arcs with $\sigma_j = \sigma$ contribute candidate points. Their parameters and squared moduli are

$$\begin{aligned} t_j(v) &= T_j(v^{\ell_*/\ell_j}), \\ z_j(v) &= \exp(w_j(t_j(v))), \\ b_j(v) &= |f(z_j(v))|^2. \end{aligned} \quad (3.11)$$

Because ℓ_*/ℓ_j is an integer, these functions are real analytic at $v = 0$. For small positive v the points are distinct: half-arcs in one window meet only at their center, and the logarithmic coordinate is injective in each window; different windows are disjoint. All nearby global maxima occur among these finitely many candidates. Each side has at least one candidate, since every nearby circle has a global maximum.

For any two applicable indices i, j , the real-analytic difference $b_i(v) - b_j(v)$ either vanishes identically near zero or has an expansion $av^n + O(v^{n+1})$ with $a \neq 0$. Its sign is therefore fixed for all sufficiently small $v > 0$. There are only finitely many such differences, so one parameter interval makes every comparison fixed. Consequently the set of indices attaining the largest squared modulus is fixed on that interval, and its cardinality is $\nu_f(e^x)$. Do this separately on both sides and exponentiate. Reducing δ so that $\delta < r_0$ proves the lemma. $\square$

Proof of Theorem 1.1. Fix $r_0 > 0$ and apply Lemma 3.4. Neither of its two constant punctured-side values can exceed $2k$, by Proposition 3.3. Hence r_0 has a punctured neighborhood containing no point of E_f . This holds at every positive radius, so E_f has no accumulation point in $(0, \infty)$ and is locally finite. Sharpness is shown in Example 4.1 below. $\square$

4. SHARPNESS AND EXCEPTIONAL CIRCLES

Example 4.1 (Sharpness). For $k \geq 1$, let

$$f_k(z) = \exp\left(z^k - \frac{z^{2k}}{4}\right).$$

Its first coefficient gap is k , since $f_k(z) = 1 + z^k + O(z^{2k})$. Set $\xi = z^k$, $|\xi| = R = r^k$, and $T = \cos(\arg \xi) \in [-1, 1]$. Using $\cos(2 \arg \xi) = 2T^2 - 1$, complete the square:

$$\begin{aligned} \log |f_k(z)| &= \operatorname{Re} \left(\xi - \frac{\xi^2}{4} \right) = RT - \frac{R^2}{4}(2T^2 - 1) \\ &= \frac{R^2}{4} + \frac{1}{2} - \frac{1}{2}(RT - 1)^2. \end{aligned}$$

For $0 < R \leq 1$, the square is minimized uniquely at $T = 1$, so the unique maximizing ξ -value is $\xi = R$. For $R > 1$, the minimum is zero at $T = 1/R$, giving exactly the two values $\xi = 1 \pm i\sqrt{R^2 - 1}$. At $R = 1$ these two values coincide. Every nonzero ξ has exactly k distinct preimages under $z \mapsto z^k$, all of modulus r . Consequently,

$$\nu_{f_k}(r) = \begin{cases} k, & 0 < r \leq 1, \\ 2k, & r > 1. \end{cases}$$

Thus the constant $2k$ is attained on an entire tail. It is also possible to see the degree bound directly in this example. Here $q(z) = kz^k - \frac{k}{2}z^{2k}$ has real coefficients, so $q^* = q$. Writing $u = z^k$ and $v = w^k$, the correspondence factors as

$$q(z) - q(w) = k(u - v) \left(1 - \frac{u + v}{2} \right).$$

On the part $u + v = 2$, one has $uv = s^k$ and $p = k(u - u^2/2) = \frac{k}{2}s^k$. For generic s , the equation $u(2 - u) = s^k$ has two distinct nonzero roots. Each root has k choices of z , after which $w = s/z$ is determined. The other factor $u = v$ meets this image graph only when $u = v = 1$ and $s^k = 1$, so it contributes no additional generic pairs. Hence this image component has generic degree $2k$, exactly the bound of Theorem 2.6.

Example 4.2 (Large multiplicity at an exceptional radius). Fix $N \geq 3$ and $0 < \tau < 1/8$, and put

$$F(z) = \exp \left(z^N + \tau z(1 - z^N)^4 \right).$$

Then $F(0) = 1$, $F'(0) = \tau$, and the first coefficient gap is one. Nevertheless, $\nu_F(1) = N$. To see this, let $d = |1 - z^N|$ on $|z| = 1$. Since $|z^N| = 1$, one has $1 - \operatorname{Re}(z^N) = |1 - z^N|^2/2 = d^2/2$. Also, $|\tau z(1 - z^N)^4| = \tau d^4$ and $d \leq 2$, so

$$1 - \operatorname{Re}(z^N + \tau z(1 - z^N)^4) \geq \frac{d^2}{2} - \tau d^4 \geq \left(\frac{1}{2} - 4\tau \right) d^2.$$

The coefficient $\frac{1}{2} - 4\tau$ is positive. Thus the real part of the exponent is strictly below 1 when $z^N \neq 1$, and equals 1 at every N th root of unity. These are precisely the N maximum-modulus points. Differentiating the exponent gives the logarithmic derivative

$$q(z) = Nz^N + \tau z(1 - z^N)^4 - 4N\tau z^{N+1}(1 - z^N)^3.$$

Hence $F(\zeta) = e$ and $q(\zeta) = N$ for $\zeta^N = 1$. The fiber over $(s, p) = (1, N)$ therefore contains at least N distinct physical pairs $(\zeta, \bar{\zeta})$.

This target point is singular in the total reduced image Y : at a smooth point, Lemma 2.4 and Theorem 2.6 would bound the fiber by two. Several local image branches can meet at a singular point even when they belong to the same global irreducible component. This is why the proof removes singular image points before counting fibers.

Theorem 1.1 bounds persistent multiplicity, not every singular fiber. The examples above exhibit this distinction directly. In particular, large values on isolated circles are compatible with the theorem, whereas a lower bound tending to infinity on every sufficiently large circle is not.

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